

PHYSICS**Paper 1 (Theory)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 8 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given below these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1. Planck's constant	h	$= 6.6 \times 10^{-34} \text{ J s}$
2. Speed of light in vacuum	c	$= 3 \times 10^8 \text{ m s}^{-1}$
3. Charge of a proton	e	$= 1.6 \times 10^{-19} \text{ C}$
4. Mass of a proton	m_p	$= 1.67 \times 10^{-27} \text{ kg}$
5.	$1/4\pi\epsilon_0$	$= 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
6. Permeability of vacuum	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
7. One atomic mass unit	1 u	$= 1.66 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$
8. One electron volt	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
9.	1 MeV	$= 1.6 \times 10^{-13} \text{ J}$

Note: The questions in this sample paper follow the format of the ISC Specimen Question Paper 2027. The weightage allocated to each unit in the syllabus has been strictly adhered to.

SECTION A – 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

(i) A Wheatstone bridge ABCD has $P = 2\ \Omega$ in arm AB, $Q = 4\ \Omega$ in arm BC, $R = 6\ \Omega$ in arm AD and an unknown S in arm DC; the galvanometer is connected between B and D. If the bridge is balanced, the value of S is: (*Apply*) [1]

- (a) $3\ \Omega$
- (b) $8\ \Omega$
- (c) $12\ \Omega$
- (d) $1.33\ \Omega$

(ii) Match the entries in Column I with those in Column II. (*Recall*) [1]

Column I	Column II
1. Diamagnetic	A. Susceptibility large and positive; shows hysteresis
2. Paramagnetic	B. Temperature above which a ferromagnet becomes paramagnetic
3. Ferromagnetic	C. Susceptibility small, negative and independent of temperature
4. Curie temperature	D. Susceptibility small, positive and inversely proportional to temperature

- (a) 1-C, 2-D, 3-A, 4-B
- (b) 1-D, 2-C, 3-A, 4-B
- (c) 1-C, 2-A, 3-D, 4-B
- (d) 1-A, 2-D, 3-B, 4-C

(iii) A proton moving along the $+x$ direction enters a uniform magnetic field directed along the $+y$ direction. The magnetic force on it at that instant is along: (*Apply*) [1]

- (a) $+x$
- (b) $-y$
- (c) $+z$
- (d) $-z$

(iv) Identify the statements that are correct for the photoelectric effect. (*Understand*) [1]

- (P) The maximum kinetic energy of the photoelectrons increases linearly with the frequency of the incident light.
- (Q) The photocurrent is proportional to the intensity of the incident light (above the threshold).
- (R) The stopping potential increases when the intensity is increased.
- (S) Emission occurs for any frequency if the light is intense enough.

- (a) Both (P) and (Q)
- (b) Both (Q) and (R)
- (c) Both (R) and (S)
- (d) Both (P) and (S)

- (v) Select the odd one out from the following materials. (*Analyse*) [1]
- (a) Silicon
 - (b) Germanium
 - (c) Gallium arsenide
 - (d) Copper
- (vi) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]
- Assertion:** Diffraction of sound is observed in everyday life far more commonly than diffraction of light.
- Reason:** Diffraction is appreciable only when the size of the obstacle or aperture is comparable to the wavelength, and the wavelength of light is very small. (*Analyse*)
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Assertion is false and Reason is true.
- (vii) In a compound microscope, the image formed by the objective is: (*Recall*) [1]
- (a) virtual, erect and magnified
 - (b) real, inverted and magnified
 - (c) real, erect and diminished
 - (d) virtual, inverted and diminished
- (B) Answer the following questions briefly:
- (i) A hollow conducting sphere carries a charge Q on its surface. What is the electric field at its centre? (*Recall*) [1]
- (ii) The frequency of the a.c. supply across a capacitor is doubled. What happens to its capacitive reactance? (*Understand*) [1]
- (iii) Fill in the blank with an appropriate term from the bracket. [1]
In Young's double slit experiment, the fringe width is _____ proportional to the separation between the slits. (directly, inversely, not) (*Recall*)
- (iv) An electron and a proton have the same kinetic energy. Which of them has the longer de Broglie wavelength? (*Understand*) [1]
- (v) Why are the control rods of a nuclear reactor made of cadmium or boron? (*Recall*) [1]
- (vi) What is meant by **doping** a semiconductor? (*Recall*) [1]
- (vii) Find the angle of minimum deviation for an equilateral glass prism of refractive index $\sqrt{2}$. (*Apply*) [1]

SECTION B – 14 MARKS

Question 2

- (i) Three point charges $+2\ \mu\text{C}$, $-3\ \mu\text{C}$ and $+4\ \mu\text{C}$ are placed at the corners of an equilateral triangle of side $0.3\ \text{m}$ in air. Calculate the electrostatic potential energy of the system. [2]
(Apply)

OR

- (ii) A parallel plate air capacitor of capacitance $10\ \text{pF}$ is charged to $100\ \text{V}$ and then disconnected [2]
from the battery. A slab of dielectric constant 5 is now made to fill the space between the plates completely. Find the new capacitance, the new potential difference and the new energy stored. (Apply)

Question 3

Figure 1 shows how the terminal potential difference V of a cell varies with the current I drawn from it.

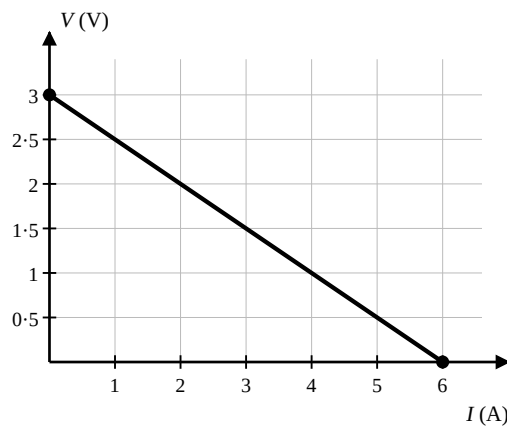


Figure 1

- (i) Find the emf and the internal resistance of the cell. (Analyse) [1]
(ii) What is the terminal potential difference when a current of $1.0\ \text{A}$ is drawn? (Apply) [1]

Question 4

- (i) In a potentiometer, a cell of emf $1.5\ \text{V}$ is balanced at a length of $60\ \text{cm}$ of the wire. When it [2]
is replaced by another cell, the balance length becomes $80\ \text{cm}$. Find the emf of the second cell. Why is a potentiometer preferred to a voltmeter for measuring emf? (Apply)

OR

- (ii) Two bulbs rated $100\ \text{W}$, $220\ \text{V}$ and $60\ \text{W}$, $220\ \text{V}$ are connected in series to a $220\ \text{V}$ supply. [2]
Which bulb glows brighter? Justify your answer. (Analyse)

Question 5

- The current in a coil falls steadily from $6\ \text{A}$ to zero in $0.03\ \text{s}$, and an average emf of $120\ \text{V}$ is [2]
induced in it. Calculate the self-inductance of the coil and the energy that was stored in it when the current was $6\ \text{A}$. (Apply)

Question 6

- (i) Name the electromagnetic waves (a) used to detect fractures in bones, and (b) used in radar systems that locate aircraft. (*Recall*) [1]
- (ii) An FM radio station broadcasts at 90 MHz. Calculate the wavelength of its waves. (*Apply*) [1]

Question 7

- (i) What is meant by the depletion region of a p-n junction? (*Recall*) [1]
- (ii) How is the potential barrier across the junction formed? (*Understand*) [1]

Question 8

An electron and an α -particle have the same de Broglie wavelength. Compare (a) their momenta and (b) their kinetic energies. (*Analyse*) [2]

SECTION C – 27 MARKS**Question 9**

- (i) Derive an expression for the electric field intensity at a point on the equatorial line of an electric dipole of moment $p = q \times 2l$, at a distance r from its centre. (*Understand*) [3]

OR

- (ii) Using Gauss' theorem, obtain an expression for the electric field intensity due to an infinite plane sheet of charge of uniform surface charge density σ . Draw the Gaussian surface you use. (*Understand*) [3]

Question 10

- (i) In the potentiometer circuit of Figure 2, the driver cell has an emf of 6.0 V and negligible internal resistance, and the wire AB is 4.0 m long with a resistance of 20 Ω . Calculate (a) the current in AB, (b) the potential gradient along AB, and (c) the balance length for a cell of emf 1.2 V. (*Apply*) [3]

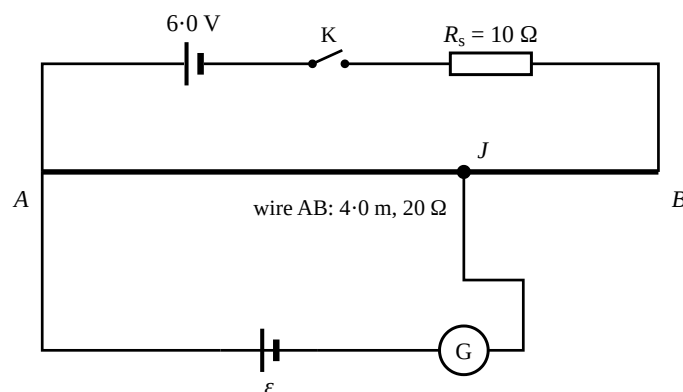


Figure 2

OR

- (ii) Twelve identical cells, each of emf 1.5 V and internal resistance 0.5 Ω , are arranged in three parallel rows of four cells each in series. The battery is connected across a 2 Ω resistor. Calculate (a) the current in the resistor, (b) the current through each cell, and (c) the terminal voltage of the battery. (*Apply*) [3]

Question 11

A beam of protons moving with velocity $\mathbf{v}$ enters a region of uniform magnetic field $B = 0.02 \text{ T}$ directed into the plane of the paper, as shown in Figure 3. A uniform electric field of $3.0 \times 10^4 \text{ V m}^{-1}$ is applied in the same region so that the beam passes undeviated.

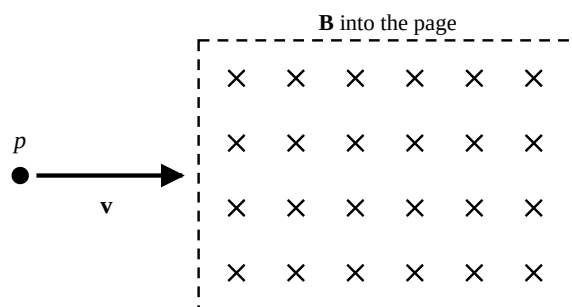


Figure 3

- (i) What should be the direction of the electric field? (*Analyse*) [1]
- (ii) Calculate the speed of the protons. (*Apply*) [1]
- (iii) If the electric field is switched off, find the radius of the circular path of the protons. (*Apply*) [1]

Question 12

- (i) A rectangular coil of 50 turns, $8 \text{ cm} \times 5 \text{ cm}$, carries a current of 2 A in a uniform magnetic field of 0.5 T . The normal to the plane of the coil makes an angle of 30° with the field. Calculate the torque on the coil. In which orientation is the torque zero? (*Apply*) [2]
- (ii) The north pole of a bar magnet is moved towards a closed coil, as shown in Figure 4. What is the direction of the induced current in the coil as seen from the magnet? (*Analyse*) [1]

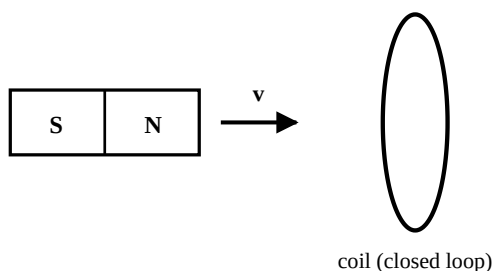


Figure 4

Question 13

A physics teacher asked the following questions on lenses and mirrors. State whether the answer of the student is correct or incorrect. Correct the incorrect response with a brief explanation. (*Evaluate*)

- (i) **Teacher:** An object is placed 30 cm from a convex lens of focal length 15 cm . Where is the image formed? [1]
Student: At 30 cm on the other side of the lens; real, inverted and of the same size.
- (ii) **Teacher:** What happens to the image if the lower half of the lens is covered with black paper? [1]
Student: Only the upper half of the image is formed.

- (iii) **Teacher:** A concave mirror of focal length 20 cm is immersed in water ($n = 4/3$). What is its focal length now? [1]

Student: It becomes $20 \times 4/3 = 26.7$ cm.

Question 14

- In Young's double slit experiment, the slits S_1 and S_2 are separated by d and the screen is at a distance D ($D \gg d$). With the help of a diagram, obtain an expression for the fringe width. [3]
(Understand)

Question 15

- Using Huygens' principle, draw the reflected wavefront when a plane wavefront is incident obliquely on a plane reflecting surface. Hence prove the law of reflection. (Understand) [3]

Question 16

Figure 5 shows the variation of photocurrent with the anode potential for light of the same frequency but two different intensities, curves 1 and 2.

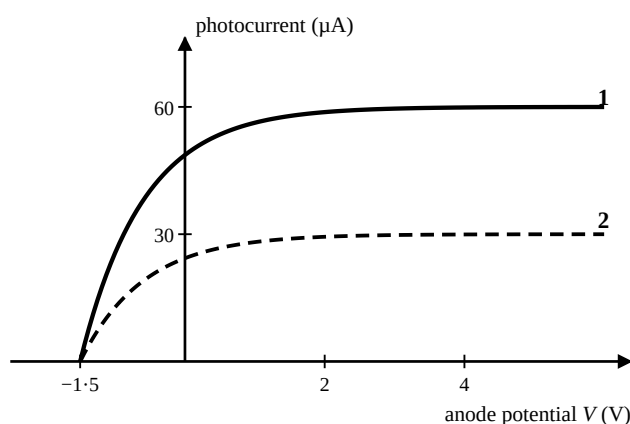


Figure 5

- (i) Which curve corresponds to the higher intensity? Why? (Analyse) [1]
(ii) Why do both curves meet the potential axis at the same point? (Understand) [1]
(iii) The incident light has a wavelength of 330 nm. Find the work function of the cathode in eV. [1]
(Apply)

Question 17

- (i) Draw a labelled circuit diagram of a full-wave rectifier using two junction diodes. Show the input and output waveforms. (Understand) [2]
(ii) If the input a.c. has a frequency of 50 Hz, what is the frequency of the output? (Apply) [1]

SECTION D – 15 MARKS

Question 18

- (i)(a) A coil draws a current of 2.0 A from a 100 V d.c. supply, and 1.0 A from a 100 V, 50 Hz a.c. supply. Calculate its resistance, its self-inductance and its power factor on the a.c. supply. [3]
(Apply)

- (b) A metal rod of length 0.5 m slides on a pair of horizontal rails in a vertical magnetic field of 0.4 T. Its displacement is $x = (2t^2 + 3t)$ m. Calculate the emf induced across the rod at $t = 2$ s. (Apply) [2]

OR

- (ii) A series LCR circuit with $L = 0.5$ H, $C = 8 \mu\text{F}$ and $R = 25 \Omega$ is connected to a 200 V a.c. source of variable frequency.
- (a) Find the resonant angular frequency and the resonant frequency. (Apply) [1]
- (b) Find the rms current at resonance. (Apply) [1]
- (c) Find the rms voltage across the inductor at resonance. Why does it not appear in the total? (Analyse) [1]
- (d) Sketch graphs of rms current against frequency for this circuit and for the same circuit with a larger resistance. Which gives the sharper resonance? (Understand) [2]

Question 19

- (i)(a) The radius of a nucleus of mass number A is $R = R_0 A^{1/3}$. Show that the density of nuclear matter is independent of A . (Apply) [2]
- (b) Calculate the energy released in the fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + {}^1_0\text{n}$. (Masses: ${}^2_1\text{H} = 2.014102$ u, ${}^3_2\text{He} = 3.016029$ u, $\text{n} = 1.008665$ u) (Apply) [2]
- (c) Why is fusion called a thermonuclear reaction? (Understand) [1]

OR

- (ii)(a) Using Bohr's postulates, obtain an expression for the total energy of the electron in the n^{th} orbit of a hydrogen atom and show that $E_n \propto -1/n^2$. (Apply) [3]
- (b) Prove that $1 \text{ u} = 931.5 \text{ MeV}$. (Apply) [2]

Question 20

An optics kit for a school exhibition contains a thin convex lens L_1 of focal length 20 cm, a thin concave lens L_2 of focal length 30 cm, both made of glass of refractive index 1.5, and a concave mirror M of radius of curvature 40 cm. The students use them to build a slide projector, a magnifier and a 'light collector' for a small telescope.

- (i) L_1 and L_2 are placed in contact. Find the focal length and the nature of the combination. (Apply) [1]
- (ii) Using L_1 alone, a slide is to be projected sharply on a screen 80 cm from the lens. How far from the lens must the slide be placed, and what is the magnification? (Apply) [1]
- (iii) An object is placed 30 cm in front of M . Draw a ray diagram to show the formation of the image and state its nature. (Understand) [1]
- (iv) What will be the focal length of L_1 when it is immersed in water ($n = 4/3$)? (Apply) [1]
- (v) Why would M be preferred to L_1 as the light-collecting element of a telescope? (Recall) [1]

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